

Chapter 95 — Urinary Lithiasis: Etiology, Epidemiology, and Pathogenesis — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Calcium in ≈ 80% of stones;** calcium oxalate 60%, uric acid 7%, struvite 7%, cystine 1–3%.
- **Commonest abnormality** in calcium formers → **hypercalciuria** (35–65%).
- **Hypercalciuria** > 200 mg/day; **hyperoxaluria** > 40; **hyperuricosuria** > 600; **hypocitraturia** < 320 mg/day.
- **Uric acid stones: low urine pH** (< 5.5) is the key determinant — uric acid excretion usually normal.
- **Calcium and oxalate contribute equally** to calcium oxalate saturation.
- **Randall plaque = calcium apatite**, origin basement membrane of thin loop of Henle.
- **Distal (type 1) renal tubular acidosis (RTA)** → calcium phosphate stones, urine pH > 6.0, profound hypocitraturia.
- **Enteric hyperoxaluria needs an intact colon.**
- **Struvite requires urease;** commonest organism **Proteus mirabilis**.
- **Borghesi 1996 RCT:** high fluid → recurrence 12% vs 27%.

EPIDEMIOLOGY

- **Lifetime prevalence 1–15%;** North America 7–13%, Europe 5–9%, Asia 1–5%.
- **US prevalence ≈ 9–10%** (NHANES, National Health and Nutrition Examination Survey); **annual incidence ≈ 2%;** rise partly **CT (computed tomography) detection**.
- **Females:** more stone morbidity (infection, nephrostomy, intensive care).
- **Hot occupations** → low volume, low pH → **uric acid stones;** sedentary → bone turnover, hypercalciuria.
- **Hard water:** not proven harmful; net probably protective.
- **Male:female** historically 2–3:1 → **1.3;** gap narrowing (female rise).
- **Race:** White > Hispanic > Black > Asian; **White M:F 2.3, Black 0.65.**
- **Adult peak** 4th–6th decades; **female 2nd peak** 6th decade (menopause).
- **Paediatric rise 5–10%/year;** peak **adolescent females.**
- **Stone belt = US Southeast;** summer peak July–September; lag < 10 days after > 10 °C.
- **Desert deployment** → stones at ≈ 93 days.
- **Energy intake > 2200 kcal/day,** BMI (body mass index), inactivity → ↑ risk.
- **Diabetes** → stones: females only; stones → diabetes: both sexes.
- **Stones ↔ CKD (chronic kidney disease), MI (myocardial infarction), stroke:** association, female-driven, not causal.

PHYSICOCHEMISTRY AND PATHOGENESIS

- **States:** < K_{sp} (solubility product) undersaturated → metastable → > K_f (formation product) unstable.
- **Heterogeneous nucleation** needs less energy than homogeneous — usual in vivo.
- **Therapy targets metastable zone** (most urinary salts sit there).
- **RSR (relative saturation ratio)** < 1 dissolve; > 1 grow.
- **JESS more reliable than EQUIL 2;** EQUIL 2 overestimates (brushite ≈ 80%, calcium oxalate ≈ 50%).
- **Oxalic acid pK ≈ 4** → pH cannot alter oxalate.
- **Calcium oxalate:** normal urine 4× solubility; precipitates at 7–11×.
- **Nephron transit 5–7 min.**
- **Randall 1937:** plaque 17% autopsies; endoscopy 74% formers vs 43% controls.
- **Ductal plugs** → suspect secondary cause (bypass, distal RTA, hyperparathyroidism, cystinuria).

- **Most abundant urinary protein:** Tamm-Horsfall (uromodulin) — inhibitor in alkaline, promoter in acid urine.
- **Citrate:** weak on calcium oxalate growth, potent on calcium phosphate growth.
- **No urinary inhibitor** of uric acid or cystine.
- **Nephrocalcin:** formers excrete non-inhibitory isoforms (lack γ-carboxyglutamate).
- **Matrix ≈ 2.5%** of stone weight.

MINERAL METABOLISM

- **Calcium absorbed 30–40%;** oxalate only **6–14%.**
- **Most potent gut calcium stimulus** → 1,25(OH)₂D₃ (calcitriol).
- **PTH (parathyroid hormone):** ↑ 1α-hydroxylase; **no direct gut action.**
- **Renal calcium:** excreted 1–3%; proximal 60–65%, TAL (thick ascending limb) 25–30%, DCT 8–10% (TRPV5).
- **Phosphate:** gut absorbs ≈ 60%; kidney reabsorbs 80–90%; PTH inhibits reabsorption.
- **Oxalobacter formigenes:** oxalate sole carbon source; ↓ with antibiotics.
- **Ascorbate** = largest endogenous oxalate source (≤ 40%); hydroxyproline ≈ 15%.

PATHOPHYSIOLOGY OF UPPER TRACT CALCULI

- **Hypercalciuria definitions:** > 200 mg/day on 400 mg calcium/100 mg sodium diet; > 4 mg/kg/day; > 7 (M) / 6 (F) mmol/day.
- **AH (absorptive hypercalciuria):** load ≥ 0.2 mg/mg creatinine, fasting < 0.11 mg/dL GF (glomerular filtrate), PTH suppressed; ≈ 30%.
- **Renal hypercalciuria:** fasting ≥ 0.11 mg/dL GF + ↑ PTH, normal serum calcium.
- **AH type II** corrects with calcium restriction; type I does not.
- **Renal phosphate leak** ≤ 2–4% (HHRH, hereditary hypophosphataemic rickets with hypercalciuria).
- **Dent disease:** X-linked, CLC-5 — hypercalciuria, proteinuria, renal failure.
- **CYP24A1 loss** → high calcitriol; idiopathic infantile hypercalcaemia.
- **PHPT (primary hyperparathyroidism):** ≈ 5% of stones; stones in < 5% of PHPT; suspect if calcium > 10.1 mg/dL; **thiazide challenge** unmasks.
- **Hypercalcaemia:** outpatient → PHPT; inpatient → malignancy.
- **Sarcoid:** PTH suppressed; corrects with steroids.
- **Primary hyperoxaluria** ≥ 100 mg/day; **PH3** severe hypercalciuria, rarely ESRD.
- **Nephrocalcinosis in PH** → ≈ 70% ↑ ESRD risk.

- **PH1:** AGT (alanine-glyoxylate aminotransferase), peroxisome, ESRD (end-stage renal disease) 2nd–3rd decade; **pyridoxine-responsive** Gly170Arg.
- **PH1 with ESRD → combined liver-kidney transplant;** PH2 → kidney alone.
- **Screen relatives for PH** before living-related donation.
- **Hyperoxaluria after bypass:** malabsorptive only, ≥ 6 months post-op.
- **Dietary oxalate:** nuts, chocolate, tea, spinach, potatoes, beets, rhubarb.
- **Hypocitraturia causes:** distal RTA, diarrhoea, animal protein, thiazides, ACE (angiotensin-converting enzyme) inhibitors.
- **Ettinger 1986 RCT:** allopurinol > 2× recurrence reduction (hyperuricosuric, normocalciuric).
- **Confirm distal RTA:** ammonium chloride load, pH not < 5.5; stones ≤ 70%.
- **Incomplete distal RTA:** no acidosis, normal electrolytes.
- **SLC4A1** distal RTA: dominant, hearing spared; H⁺-ATPase forms recessive.
- **Proximal RTA:** HCO₃⁻ 15–18 mEq/L, urine pH < 5.5, no stones.
- **Type 4 RTA:** hyperkalaemic, can acidify; uric acid if any.
- **Uric acid pKa 5.35;** solubility ≈ 100 mg/L; > 1200 mg/L soluble at pH 6.5.
- **BMI ≥ 27** → urine pH ≤ 5.5; type 2 diabetes → uric acid stones ↑.
- **Insulin resistance** → ↓ ammoniogenesis → low pH.
- **Cystine solubility** 300 / 400 / 1000 mg/L at pH 5 / 7 / 9.
- **Cystinuria:** autosomal recessive; FE (fractional excretion) 0.4% → 100%; ≤ 10% paediatric stones; first stone ≈ 12 years.
- **ICC (International Cystinuria Consortium):** A = SLC3A1 (chr 2), B = SLC7A9 (chr 19).
- **Struvite urine pH 7.2–8.0;** F:M 2:1; highest staghorn risk → complete cord transection.
- **Staghorns:** majority metabolic (calcium phosphate), not struvite.
- **Xanthine, DHA (2,8-dihydroxyadenine):** radiolucent; DHA → febuxostat > allopurinol.
- **Ammonium acid urate** ≈ 1%; pH 6.2–6.3; ileostomy, laxatives, obesity.
- **Indinavir:** stones 4–13%; ≈ 50% excreted unchanged; soluble in acid urine.
- **Atazanavir–ritonavir** ≈ 7% stones; triamterene ≈ 0.4% of stones.
- **Carbonic anhydrase inhibitors** → calcium phosphate stones.
- **Topiramate:** label 1.2–1.5% underestimates; ≈ 20% asymptomatic on CT.
- **Furosemide:** stones ≤ 64% low-birth-weight infants.
- **Anatomy:** UPJO (ureteropelvic junction obstruction) ≈ 20%; horseshoe 36% adults; diverticulum ≤ 40%; horseshoe → hypocitraturia over-represented.
- **MSK (medullary sponge kidney):** hypercalciuria, hypocitraturia, distal RTA.
- **Pregnancy:** 1:204–1:3000; hydronephrosis ≤ 90%, right > left; GFR (glomerular filtration rate) +30–50%; misdiagnosis ≤ 28%.
- **Pregnancy peak** 0–3 months post partum; stones mainly calcium phosphate.

CLASSIC TRAPS

- **Trap:** uric acid stones = hyperuricosuria — **Actually:** low urine pH.
- **Trap:** oxalate outweighs calcium — **Actually:** equal contributors.
- **Trap:** Randall plaque is calcium oxalate — **Actually:** calcium apatite.
- **Trap:** ileostomy → hyperoxaluria — **Actually:** no colon, uric acid/ammonium urate.
- **Trap:** restrict calcium to cut oxalate — **Actually:** raises oxalate absorption.
- **Trap:** proximal RTA forms stones — **Actually:** citrate normal; distal does.
- **Trap:** E. coli makes urease — **Actually:** rarely; cultures discordant.
- **Trap:** fixing UPJO stops stones — **Actually:** metabolic recurrence, often contralateral.
- **Trap:** Bartter = acidosis — **Actually:** hypokalaemic alkalosis.
- **Trap:** restrictive bariatric surgery → stones — **Actually:** only malabsorptive.